

Maximum Nullity of Cyclic Covers of Graphs, and Exact Zero Forcing Numbers of Generalized Petersen Graphs

Aryan Padarthy

September 23, 2026

Abstract

For a graph G let $Z(G)$ be its zero forcing number and $M(G)$ the maximum nullity over matrices whose off-diagonal support is exactly $E(G)$, so that $M(G) \leq Z(G)$. We study both for *cyclic covers*: given a base graph B with edge voltages in $\mathbb{Z}_n$, the derived graph B^n on $V(B) \times \mathbb{Z}_n$. Generalized Petersen graphs are the case of a two-vertex base with voltages 1, k and 0.

Our main structural result is that for such a cover the nullity is governed by a single scalar object determined by the voltages. For $P(n, k)$ we prove it for *every* matrix carrying the pattern, symmetric or not: eliminating the inner coordinates leaves one nine-term linear recurrence of order $2k + 2$, so

$$\text{null } A = \dim \ker(T - I) \leq 2k + 2,$$

T the monodromy, with equality iff $T = I$. The order $2k + 2$ is exactly the size of the forcing set produced by a rotation-bootstrap argument that gives $Z(P(n, k)) \leq 2k + 2$, so *the ceiling of the matrix-certificate method coincides with the upper bound it is trying to match*: either the method settles Z completely or nothing does. For a general cover the same reduction holds for equivariant matrices, with the ceiling given by the degree span of $\det M(\zeta)$.

We then reach the ceiling. Under the rotation the singularity condition is a monic degree- $(k + 1)$ polynomial in $s = 2 \cos(2\pi m/n)$ carrying exactly three free parameters for every k ; three roots may be prescribed and no more, but a symbol with rational coefficients drags its whole Galois orbit onto the grid $\{2 \cos(2\pi m/n)\}$. Choosing the symbol to be a product of minimal polynomials of $2 \cos(2\pi/d)$ yields explicit *integer* certificates and hence

$$Z(P(n, 4)) = M(P(n, 4)) = 10, \quad Z(P(n, 5)) = M(P(n, 5)) = 12,$$

for every n divisible by 60, 70 or 90, respectively by 24 — the first exact values of $Z(P(n, k))$ for $k \geq 4$, a range a 2026 correction note records as open. In the second case the certificate is the adjacency matrix itself, and we classify its nullity completely: with $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$, $\text{null}(\text{Adj } P(n, k)) = \sum \varphi(d)$ over those d with $\Phi_d \mid P_k$ and $d \mid n$, a Conway–Jones phenomenon; in particular the adjacency matrix is never singular for $k = 3, 6, 9$.

Against this we prove a matching *obstruction*. Realisability of a certificate is the positivity (or nonvanishing) of $c^2 = a\alpha - \beta$, and because $L_k(-s) = (-1)^k L_k(s)$, for *every even* k a symbol with two roots $y, -y$ has $c^2 = 0$ identically, while for odd k no such obstruction exists. For $k = 2$, where the symbol has three roots and no residual conditions, this is the whole story in closed form:

$$c^2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3),$$

so the construction fails precisely when two prescribed roots are antipodal. Counting antipodal classes on the grid then settles $k = 2$ for all n : $Z(P(n, 2)) = M(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, by an explicit certificate at the three most negative grid values; for $7 \mid n$ no symmetric certificate exists and an integer non-symmetric one of nullity 6 does, the one place where the larger matrix class is needed rather than merely available. The two exceptions are of

opposite kinds — $Z(P(8, 2)) = 5$, so there the method is right to fail, and $Z(P(10, 2)) = 6$, so there it is the method that falls short.

The same analysis is what settles the remaining cases, because only $c^2 = 0$ is fatal: a symbol with $c^2 < 0$ admits no symmetric certificate but is realisable once the two spoke weights are allowed to differ. Redoing our exhaustive enumeration of rational symbols under the weaker condition turns two cases we had recorded as impossible into exact values. Both come from the *signed* adjacency matrix, the adjacency matrix of $P(n, k)$ with one set of spokes negated, whose nullity we classify by the polynomial $Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1$ — the unsigned P_k with the middle sign reversed. Q_k splits into cyclotomics only for $k = 2, 3, 7$ in $2 \leq k \leq 60$, and at $k = 3, 7$ it gives the ceiling:

$$Z(P(n, 3)) = 8 \quad (10 \mid n), \quad Z(P(n, 7)) = 16 \quad (120 \mid n),$$

by matrices of zeros and ± 1 , the second replacing a floating-point claim by an exact one. Conversely $k = 6, 8, 9, 10$ are *impossible* over $\mathbb{Q}$ in both classes, and now for an identified reason: every candidate clearing the family condition is one of at most 25 degenerate symbols, the bound coming from Niven’s theorem.

Breaking the symmetry to period d , the map from weights to the coefficients of the symbol has rank $\min(2d + 1, k + 1)$ in every case computed, so every symbol becomes reachable exactly when $d \geq \lceil k/2 \rceil$. This yields $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$, $Z(P(96, 7)) = 16$ and $Z(P(96, 8)) = 18$, though these certificates are verified numerically rather than in exact arithmetic. We state throughout which claims are proved, which are computations, and which earlier claims of ours are corrected.

1 Background

Given a set S of *filled* vertices of a graph G , the colour change rule says: if a filled vertex has exactly one unfilled neighbour, that neighbour becomes filled. Write $\text{cl}(S)$ for the closure of S under repeated application. S is a *zero forcing set* if $\text{cl}(S) = V$, and $Z(G)$ is the least size of one. $Z(G)$ bounds the maximum nullity over the symmetric matrices with G ’s off-diagonal pattern, and arose independently in the control of quantum systems and in sensor placement for power networks.

The families we treat all have $2n$ or $4n$ vertices, are cubic, and carry the rotation $\rho : i \mapsto i + 1$ acting on index subscripts.

- $P(n, k)$: outer $u_i u_{i+1}$, inner $v_i v_{i+k}$, spokes $u_i v_i$.
- $I(n, j, k)$: outer $u_i u_{i+j}$, inner $v_i v_{i+k}$, spokes $u_i v_i$; thus $P(n, k) = I(n, 1, k)$.
- $DP(n, k)$: two outer cycles $u_i u_{i+1}$ and $x_i x_{i+1}$, inner edges $w_i y_{i+k}$ and $y_i w_{i+k}$, spokes $u_i w_i$ and $x_i y_i$.

Rashidi, Shajareh Poursalavati and Tavakkoli [1] proved $Z(P(n, k)) \leq 2k+2$ and $Z(P(n, 2)) = 6$ for $n \geq 10$. Their Theorem 3.6, claiming $Z(P(n, 3)) = 8$ for $n \geq 12$, is false: Krishnan [2] exhibits $Z(P(12, 3)) = 7$, the flaw being a case analysis that never excludes 7-vertex sets. That note closes by recording that determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.

2 The rotation bootstrap

The following two lemmas are elementary; the point is that together they reduce an “all n ” statement to one finite check.

Lemma 1 (Bootstrap). *Let ρ be an automorphism of a finite graph G and let $S \subseteq V(G)$ satisfy $\rho(S) \subseteq \text{cl}(S)$. Then $\rho(\text{cl}(S)) = \text{cl}(S)$.*

Proof. Closure commutes with automorphisms, and is monotone and idempotent, so $\rho(\text{cl}(S)) = \text{cl}(\rho(S)) \subseteq \text{cl}(\text{cl}(S)) = \text{cl}(S)$. As ρ permutes the finite set $V(G)$, the two sets have equal cardinality, hence are equal. $\square$

Lemma 2 (Closing). *Let G be cubic with vertex set $O \sqcup I$ such that every vertex of I has exactly one neighbour in O , every vertex of O has exactly one neighbour in I , and the remaining neighbours of each vertex of O lie in O . If $O \subseteq \text{cl}(S)$ then $\text{cl}(S) = V(G)$.*

Proof. Suppose some $w \in I$ is unfilled and let $o \in O$ be its unique neighbour in O . The other two neighbours of o lie in O , hence are filled, so w is the unique unfilled neighbour of the filled vertex o and is forced. $\square$

To apply these to a family it suffices to exhibit one set S , described uniformly in n , and verify $\rho(S) \subseteq \text{cl}(S)$ by an explicit finite forcing chain. Since ρ 's orbit of any outer vertex is all of O , Lemmas 1 and 2 then give $\text{cl}(S) = V$ and so $Z(G) \leq |S|$, for every n in range.

3 I-graphs

Theorem 3. *For all $j, k \geq 1$ and $n \geq 2(j + k) + 1$, $Z(I(n, j, k)) \leq 2(j + k)$.*

Proof. Put $m = 2(j + k)$ and $S = \{u_0, \dots, u_{m-1}\}$, so $|S| = m$. Since $n \geq m + 1$ the indices $0, \dots, m$ are pairwise distinct mod n .

Step 1. The outer neighbours of u_i are $u_{i \pm j}$, both in S precisely when $j \leq i \leq m - 1 - j$. For such i the spoke neighbour v_i is therefore the unique unfilled neighbour of u_i and is forced. This fills $v_j, \dots, v_{m-1-j}$, that is $v_j, \dots, v_{j+2k-1}$.

Step 2. Consider v_{j+k} . Its neighbours are $u_{j+k} \in S$ (as $j + k \leq m - 1$), v_j , filled in Step 1, and v_{j+2k} , which is not: Step 1 filled indices at most $j + 2k - 1$. So v_{j+k} forces v_{j+2k} .

Step 3. Consider $u_{j+2k} = u_{m-j}$. Its neighbours are $u_{2k} \in S$ (as $2k \leq m - 1$, using $j \geq 1$), the vertex v_{j+2k} filled in Step 2, and u_m , which is unfilled. So u_{j+2k} forces u_m .

Hence $\rho(S) = \{u_1, \dots, u_m\} \subseteq \text{cl}(S)$. By Lemma 1, $\text{cl}(S)$ is ρ -invariant; it contains u_0 , whose ρ -orbit is all of $O = \{u_0, \dots, u_{n-1}\}$, so $O \subseteq \text{cl}(S)$, and Lemma 2 gives $\text{cl}(S) = V$. $\square$

Corollary 4. *For $k \geq 1$ and $n \geq 2k + 3$, $Z(P(n, k)) \leq 2k + 2$.*

Proof. $P(n, k) = I(n, 1, k)$; take $j = 1$ in Theorem 3. $\square$

Remark 5. The corollary is [1, Thm. 2.2]. We note it because that paper is known to contain a false theorem, and because the corrected argument given in [2] is written only for $k = 3$; the proof above is uniform in j and k . Every step of the chain has been verified by computer for $1 \leq j \leq 4$, $1 \leq k \leq 5$ and six values of n per pair (106 cases).

Remark 6 (Scope). Theorem 3 should not be read as $2(j + k)$ new bounds. If $\gcd(n, j) = 1$ then multiplying indices by $j^{-1} \bmod n$ is an isomorphism $I(n, j, k) \cong I(n, 1, j^{-1}k) = P(n, j^{-1}k)$, so those graphs are generalized Petersen graphs and the corollary already applies. We verified this in five cases, e.g. $I(15, 2, 4) \cong P(15, 2)$ with $Z = 6$ on both sides and $I(13, 2, 5) \cong P(13, 4)$ with $Z = 8$. The genuinely new content of Theorem 3 is therefore confined to $\gcd(n, j) > 1$ and $\gcd(n, k) > 1$.

In that regime the bound is usually far from sharp: among the cases we computed it is attained only at $I(18, 2, 3)$ ($Z = 10 = 2(2 + 3)$), while e.g. $I(20, 5, 8)$ has $Z = 10$ against a bound of 26. The

reason is structural. When $\gcd(n, j) = d > 1$ the outer vertices form d cycles of length n/d rather than one cycle, and the set $\{u_0, \dots, u_{m-1}\}$ used in the proof spreads across all of them; a smaller, better-adapted set suffices. We record the bound as proved and its sharpness in this regime as open.

4 Double generalized Petersen graphs

Theorem 7. For all $k \geq 1$ and $n \geq 2k + 3$, $Z(DP(n, k)) \leq 4k + 4$.

Proof. Put $S = \{u_0, \dots, u_{2k+1}\} \cup \{x_0, \dots, x_{2k+1}\}$, so $|S| = 4k + 4$.

Step 1. For $1 \leq i \leq 2k$ the outer neighbours $u_{i\pm 1}$ of u_i lie in S , so u_i forces w_i ; symmetrically x_i forces y_i . This fills $w_1, \dots, w_{2k}$ and $y_1, \dots, y_{2k}$.

Step 2. The neighbours of w_{k+1} are $u_{k+1} \in S$, y_1 (filled) and y_{2k+1} (not), so w_{k+1} forces y_{2k+1} . Symmetrically y_{k+1} forces w_{2k+1} .

Step 3. The neighbours of u_{2k+1} are $u_{2k} \in S$, w_{2k+1} (filled in Step 2) and u_{2k+2} , so u_{2k+1} forces u_{2k+2} ; symmetrically x_{2k+1} forces x_{2k+2} .

Hence $\rho(S) \subseteq \text{cl}(S)$. The outer set here is $O = \{u_i\} \cup \{x_i\}$, on which ρ acts with two orbits, both met by $\text{cl}(S)$; Lemmas 1 and 2 finish as before. $\square$

No zero forcing results for $DP(n, k)$ were found in the literature; the family is studied for Hamiltonicity, super-connectivity and its automorphism group. Every step above was verified by computer for $1 \leq k \leq 5$ and eight values of n per k .

5 Computed values

All values are exact, by exhaustive search over vertex subsets with a 128-bit mask closure, reduced by ρ : a minimum forcing set may be assumed to contain a fixed orbit representative. The implementation reproduced every published value before use — Krishnan’s table of $Z(P(n, 3))$ for $7 \leq n \leq 20$ (including the corrected $Z(P(12, 3)) = 7$ and the non-monotone values at $n = 10, 11$), $Z(P(n, 2)) = 6$ for $10 \leq n \leq 16$, and $Z(P(2k + 1, k)) = 6$ for $k = 5, 6, 7$ — and was independently cross-checked against a brute-force implementation and two exact solver formulations.

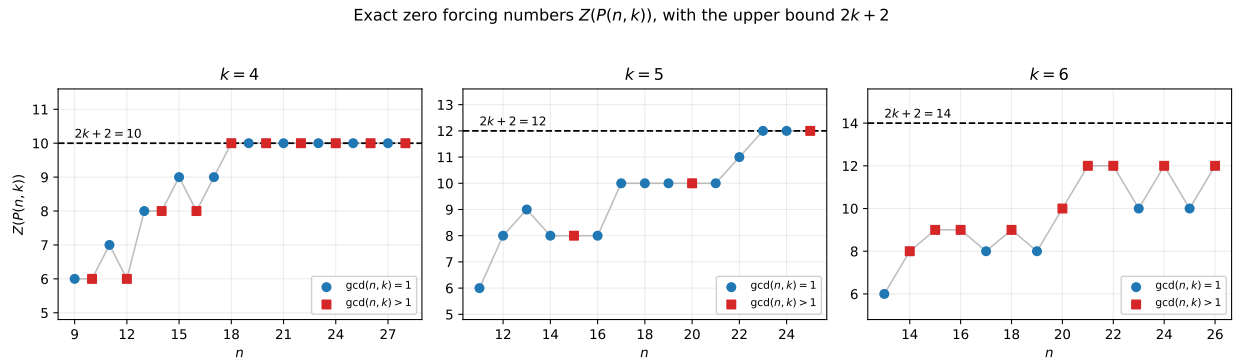


Figure 1: Exact $Z(P(n, k))$ for $k = 4, 5, 6$ against the bound $2k + 2$. Points are separated by whether $\gcd(n, k) = 1$. The sequences are non-monotone: $k = 4$ drops at $n = 12, 16$ and $k = 6$ at $n = 23, 25$.

k	bound $2k + 2$	least n attaining it	source
2	6	10	[1]
3	8	13	[2]
4	10	18	this work
5	12	23	this work
6	14	first at $n = 28$; not stable ($Z(P(29, 6)) = 12$)	this work

Table 1: Least n attaining the bound. For $k = 3, 4, 5$ these are $5k - 2$, but see Conjecture 8: at $k = 6$ the value is attained at $n = 28$ and then lost again at $n = 29$, so these are not stabilization thresholds.

Conjecture 8. *For each fixed k and each $d \mid k$, the values $Z(P(n, k))$ over n with $\gcd(n, k) = d$ are eventually constant and equal to $2k + 2$.*

An earlier form of this conjecture, asserting a single threshold $n_0(k) = 5k - 2$ beyond which $Z(P(n, k)) = 2k + 2$ for *all* n , is false. It fits $k = 3, 4, 5$, and predicted that $n = 28$ would be the first n with $Z(P(n, 6)) = 14$; that prediction, recorded before the computation finished, was borne out. But $Z(P(29, 6)) = 12$, and $29 > 28$, so 28 is not a stabilization point and the single-threshold form fails at $k = 6$.

The explanation is $\gcd(n, k)$. Sorting the $k = 6$ data by $d = \gcd(n, 6)$:

d	n	$Z(P(n, 6))$
1	13, 17, 19, 23, 25, 29	6, 8, 8, 10, 10, 12
2	14, 16, 20, 22, 26, 28	8, 9, 10, 12, 12, 14
3	15, 21, 27	9, 12, 13
6	18, 24	9, 12

Each class climbs on its own schedule: the class $d = 2$ attains 14 at $n = 28$ while the coprime class has only reached 12 by $n = 29$. At $k = 4$ and $k = 5$ all classes had converged within the computed range, which is why a single threshold appeared to fit; at $k = 6$ they separate. The thresholds 10, 13, 18, 23 for $k = 2, 3, 4, 5$ should therefore be read as the point at which the slowest class happened to arrive, not as values of a formula.

For DP , our computed values give $Z(DP(n, 1)) = 8 = 4 \cdot 1 + 4$ for $6 \leq n \leq 12$ and $Z(DP(n, 2)) = 12 = 4 \cdot 2 + 4$ for $n = 10, 11, 14, 15$, so Theorem 7 is attained at both $k = 1$ and $k = 2$: the bound is sharp there, not merely an upper estimate.

6 Towards a sharper bound for I-graphs

Theorem 3 is stated in terms of j and k , but the outer and inner edge sets of $I(n, j, k)$ decompose into $d_1 = \gcd(n, j)$ and $d_2 = \gcd(n, k)$ cycles respectively, and it is these that appear to govern Z . In the regime where Theorem 3 carries new content (Remark 6), it is markedly non-optimal: $I(24, 9, 10)$ has $Z = 10$ against $2(j + k) = 38$. The proof uses $2(j + k)$ *consecutive* outer vertices, which is wasteful exactly when j is large but $\gcd(n, j)$ is small, since consecutive indices stride across all d_1 outer cycles.

Observation 9. Among the 20 graphs we computed with $d_1, d_2 > 1$, every one satisfies $Z(I(n, j, k)) \leq 2(d_1 + d_2)$, with equality at $I(18, 2, 3)$, $I(18, 3, 8)$, $I(24, 3, 10)$ and $I(24, 9, 10)$.

This must be stated as an observation confined to that regime, and *not* as a bound in general. Taking $j = 1$ gives $d_1 = 1$, and if additionally $\gcd(n, k) = 1$ then $2(d_1 + d_2) = 4$, whereas $Z(P(n, k)) = 2k + 2$ grows without bound: $Z(P(11, 3)) = 7$ and $Z(P(17, 4)) = 9$ both exceed 4. So any bound in terms of d_1, d_2 alone is false, and the true quantity must retain a dependence on j and k . We flag this because the restricted observation is nonetheless suggestive: the four equality cases all have $2(j + k)$ slack by between 12 and 28, so Theorem 3 is certainly not optimal in the regime where it carries new content.

A supporting structural fact, found by computing minimum *bootstrapping* sets (those S with $\rho(S) \subseteq \text{cl}(S)$, which is what a uniform-in- n proof requires): for $I(18, 3, 8)$ the minimum such set has size $10 = 2(d_1 + d_2)$ against $2(j + k) = 22$, and decomposes as three consecutive vertices within each of the $d_1 = 3$ outer cycles together with one inner vertex. Three consecutive vertices in an outer cycle is exactly the configuration that lets the middle one force its spoke, so a proof adapted to the cycle decomposition rather than to consecutive indices should beat Theorem 3. We have not carried this out.

Remark 10 (Limits of verification). Sharpness of Theorem 3 can be tested only for very small (j, k) . The hypothesis $n \geq 2(j + k) + 1$ forces n upward while exhaustive search costs $\binom{2n-1}{2(j+k)-1}$, so the two constraints compound. On our hardware $(j, k) = (2, 3)$ takes minutes, $(3, 4)$ about ninety, $(2, 5)$ roughly a day, and $(3, 5)$ some 10^4 years. We therefore report sharpness at $(2, 3)$ — where $Z(I(n, 2, 3)) = 10 = 2(j + k)$ for $n = 18, 24$ — and note that the question is computationally out of reach beyond small parameters, rather than merely unfinished.

7 The exact ceiling of the certificate method

Every technique in Section 12 is combinatorial and fails. A matrix-theoretic route succeeds, and produces certificates. Before constructing any, we determine exactly how far the method can possibly go.

Let A be a real matrix indexed by $V(P(n, k))$ with $A_{xy} \neq 0$ iff $xy \in E$ for $x \neq y$, the diagonal unrestricted. We do *not* assume A symmetric: the bound $\text{null } A \leq Z(G)$ needs only that every edge carries a nonzero entry in each direction. Indeed if $Ax = 0$ and x vanishes on a set B , and $u \in B$ has exactly one neighbour $w \notin B$, then row u of $Ax = 0$ reads $A_{uu}x_u + \sum_{v \sim u} A_{uv}x_v = 0$, every term vanishes except $A_{uw}x_w$, and $A_{uw} \neq 0$; so $x_w = 0$, and induction along a forcing process gives $x = 0$ on $\text{cl}(B)$. Hence $x \mapsto x|_S$ is injective on $\ker A$ for every zero forcing set S , so

$$\text{null } A \leq Z(G). \quad (1)$$

Write $M(G)$ for the maximum nullity over symmetric such A and $M_{\text{cs}}(G)$ for the maximum over all of them, so $M \leq M_{\text{cs}} \leq Z$. Set

$$b_i = A_{u_i u_{i+1}}, \quad b'_i = A_{u_{i+1} u_i}, \quad c_i = A_{u_i v_i}, \quad c'_i = A_{v_i u_i}, \quad e_i = A_{v_i v_{i+k}}, \quad e'_i = A_{v_{i+k} v_i},$$

with diagonals a_i at u_i and d_i at v_i ; all of b, b', c, c', e, e' are nonzero.

Theorem 11 (Reduction). *Let $2k < n$. For $x \in \mathbb{R}^{\mathbb{Z}_n}$ put $(Lx)_i = -(b'_{i-1}x_{i-1} + a_i x_i + b_i x_{i+1})/c_i$. Then $x \mapsto (x, Lx)$ is a linear bijection from*

$$\left\{ x : \sum_{p=-k-1}^{k+1} \gamma_{i,p} x_{i+p} = 0 \text{ for all } i \in \mathbb{Z}_n \right\}$$

onto $\ker A$, where $\gamma_{i,p} \neq 0$ only for $p \in \{-k-1, -k, -k+1\} \cup \{-1, 0, 1\} \cup \{k-1, k, k+1\}$ — nine positions — and

$$\gamma_{i,k+1} = -\frac{e_i b_{i+k}}{c_{i+k}} \neq 0, \quad \gamma_{i,-k-1} = -\frac{e'_{i-k} b'_{i-k-1}}{c_{i-k}} \neq 0.$$

Consequently, with $T = T_{n-1} \cdots T_0 \in \text{GL}_{2k+2}(\mathbb{R})$ the monodromy of this recurrence on the state $(x_{i-k-1}, \dots, x_{i+k})$,

$$\text{null } A = \dim \ker(T - I), \quad \det T = \frac{(\prod_i b'_i)(\prod_i e'_i)}{(\prod_i b_i)(\prod_i e_i)},$$

so $\det T = 1$ whenever A is symmetric. In particular $\text{null } A \leq 2k+2$, with equality if and only if $T = I$.

Proof. Each spoke weight c_i is nonzero, so the equation of $\ker A$ at u_i solves uniquely for $y_i = (Lx)_i$, a 3-term expression in x . Substituting into the equation at v_i , namely $e'_{i-k} y_{i-k} + d_i y_i + e_i y_{i+k} + c'_i x_i = 0$, gives the displayed relation: the three terms contribute the index triples $\{i-k-1, i-k, i-k+1\}$, $\{i-1, i, i+1\}$ and $\{i+k-1, i+k, i+k+1\}$, and the two extreme coefficients are as stated, both nonzero. Conversely any x satisfying all the relations yields the kernel vector (x, Lx) . Since both end coefficients are nonzero the relation is a recurrence of order exactly $2k+2$, solvable forward and backward, so its solutions on $\mathbb{Z}$ form a $(2k+2)$ -dimensional space on which T acts, and the n -periodic ones are exactly the fixed vectors. Each step matrix T_i is the companion matrix of the relation, with determinant $\gamma_{i,-k-1}/\gamma_{i,k+1}$; taking the product over $i \in \mathbb{Z}_n$, every index set runs over a full residue system and the stated formula follows. $\square$

Corollary 12. $M(P(n, k)) \leq M_{\text{cs}}(P(n, k)) \leq 2k+2$ for $2k < n$.

Corollary 12 is not a new bound — it follows from (1) and Theorem 3 — but it locates the method precisely, and the identification is exact rather than approximate. The recurrence has order $2k+2$ because a nonzero kernel vector cannot vanish on $2k+2$ consecutive outer indices; and $\{u_0, \dots, u_{2k+1}\}$ is exactly the zero forcing set that the rotation bootstrap of Theorem 3 produces. So the order of the recurrence equals the size of that forcing set, and the ceiling of the certificate method coincides with the upper bound the method is trying to match, with no slack on either side. A certificate can prove $Z(P(n, k)) \geq 2k+2$ and, by Theorem 11, it does so precisely when the monodromy is the identity. Our earlier description of the method as “capped near 6–8” confused a property of the constructions we had tried with a property of the method.

Proposition 13 (Product form). *For symmetric A with $C = \text{diag}(c_i)$, put $U = A[O, O]$ and $S = C^{-1}A[I, I]C^{-1}$. Then $\text{null } A = \dim \ker(SU - I)$. Here U and S are symmetric with the outer-cycle and inner-cycle patterns respectively, free diagonals and nonzero prescribed off-diagonal entries, and every such pair arises. Hence $M(P(n, k))$ is the largest possible geometric multiplicity of the eigenvalue 1 in a product of two such matrices.*

Proof. $\ker A$ is cut out by $Ux + Cy = 0$ and $Cx + A[I, I]y = 0$; eliminating y gives $(C - A[I, I]C^{-1}U)x = 0$, and factoring out the invertible C turns this into $(I - SU)x = 0$. $\square$

Theorem 11 and Proposition 13 were verified computationally before use: over 63 pairs (n, k) with $2 \leq k \leq 7$ and three random matrices each (189 matrices), the nine-position sparsity, both end-coefficient formulas and the identity $\text{null } A = \dim \ker(T - I) \leq 2k+2$ hold in every case; over 30 further pairs and both matrix classes, so do the determinant formula, the product form, and the statement that no kernel vector vanishes on $u_0, \dots, u_{2k+1}$. See `verification/recurrence_order.py`.

8 Period-one certificates and cyclotomy

Take $A \in S(P(n, k))$ invariant under ρ , normalised so that the outer weight is 1:

$$U = aI + (P + P^{-1}), \quad V = dI + e(P^k + P^{-k}), \quad C = cI,$$

with $c, e \neq 0$. Diagonalising P splits A into the n blocks $M_m = \begin{pmatrix} a+s_m & c \\ c & d+et_m \end{pmatrix}$ with $s_m = 2\cos(2\pi m/n)$ and $t_m = 2\cos(2\pi km/n)$. Each singular block has nullity exactly 1, since $c \neq 0$, so $\text{null } A = \#\{m : \det M_m = 0\}$.

Proposition 14 (The symbol). *Let $L_k \in \mathbb{Z}[s]$ be the monic polynomial of degree k with $L_k(z + z^{-1}) = z^k + z^{-k}$ (so $L_1 = s$, $L_2 = s^2 - 2$, $L_k = sL_{k-1} - L_{k-2}$). Then $t_m = L_k(s_m)$ exactly, and with $\alpha = d/e$, $\beta = (ad - c^2)/e$,*

$$\det M_m = 0 \iff F(s_m) = 0, \quad F(s) = (s + a)L_k(s) + \alpha s + \beta.$$

F is monic of degree $k + 1$ and carries exactly three free parameters (a, α, β) , for every k . Writing $\Gamma_n = \{2\cos(2\pi m/n)\}$, each root of F in $\Gamma_n \cap (-2, 2)$ contributes 2 to $\text{null } A$ and each root at ± 2 contributes 1, because $m \mapsto n - m$ fixes s_m and is free elsewhere. Finally $c^2 = a\alpha - \beta$ after scaling $e = 1$, so the certificate is admissible precisely when $a\alpha - \beta > 0$.

Remark 15 (A claim of ours, corrected). We previously asserted that this period-one construction “never exceeds 6” for every $k \geq 2$, reasoning that only three roots can be prescribed. The premise is right and the conclusion is wrong. At a root r the condition is *linear*,

$$a L_k(r) + \alpha r + \beta = -r L_k(r), \tag{2}$$

so three distinct roots determine (a, α, β) and no fourth can be imposed. But prescribing three roots also *determines the other $k - 2$* , and those may lie in Γ_n of their own accord. They do exactly when arithmetic puts them there: if F has rational coefficients its roots form Galois orbits, and the Galois orbit of $2\cos(2\pi m/n)$ consists again of elements of Γ_n . A single rational F can therefore place all $k + 1$ roots on the grid at once. This raises the period-one ceiling from 6 to $2k + 2$, and makes the more elaborate period-two construction we reported unnecessary.

When realisability fails, exactly

The parameter count says what can be prescribed; realisability says what survives. The condition $c^2 = a\alpha - \beta \neq 0$ turns out to be a *factorisation* condition on the symbol, and this settles the question completely.

Theorem 16 (Realisability). *For a period-one matrix with symbol F ,*

$$c^2 = 0 \iff F = (s + a)(L_k(s) + t) \text{ for some constant } t.$$

Consequently the matrix is admissible in the combinatorially symmetric class if and only if F does not factor in this way, and in the symmetric class if and only if in addition $c^2 > 0$.

Proof. $c^2 = a\alpha - \beta$, so $c^2 = 0$ iff $\beta = a\alpha$, and then

$$F = (s + a)L_k + \alpha s + a\alpha = (s + a)L_k + \alpha(s + a) = (s + a)(L_k + \alpha).$$

Conversely if $F = (s + a')(L_k + t)$ then comparing the coefficient of s^k gives $a' = a$, and expanding gives $\alpha = t$, $\beta = at$, so $c^2 = at - at = 0$. The two admissibility statements are Proposition 14: the product of the two spoke weights is c^2 , which must be nonzero for the pattern to be that of $P(n, k)$, and must be a square for the matrix to be symmetric. $\square$

Over the rationals the degenerate symbols form a finite, explicit list, and the reason is Niven's theorem [4]: the only rational values of $2\cos(\pi q)$ with $q \in \mathbb{Q}$ are $0, \pm 1, \pm 2$.

Corollary 17 (The degenerate rational symbols). *Let $F \in \mathbb{Q}[s]$ be monic of degree $k+1$ with every root in Γ_n , and suppose $c^2 = 0$. Then $F = (s+a)(L_k+t)$ with*

$$a, t \in \{0, \pm 1, \pm 2\}.$$

There are therefore at most 25 degenerate rational symbols for each k .

Proof. By Theorem 16, $F = (s+a)(L_k+t)$, and a is the coefficient of s^k in F , hence rational; t is then rational too. Now $-a$ is a root of F , so $-a = 2\cos\theta$ for some rational multiple θ of 2π , and a rational forces $a \in \{0, \pm 1, \pm 2\}$ by Niven's theorem. Likewise any root r of L_k+t lies in Γ_n , so $r = 2\cos\varphi$ with φ a rational multiple of 2π and $t = -L_k(r) = -2\cos(k\varphi)$, again rational, so $t \in \{0, \pm 1, \pm 2\}$. $\square$

Enumerating the 25 pairs and keeping those for which $(s+a)(L_k+t)$ splits into distinct Ψ_d with every root interior gives 7, 8, 7, 6, 9 degenerate symbols for $k = 2, \dots, 6$, agreeing exactly with the count of $c^2 = 0$ cases found by the independent exhaustive enumeration in each case (`verification/cs_exhaustive.py`).

One special case of Theorem 16 is worth isolating, because it is visible on the root set alone and depends only on the parity of k .

Corollary 18 (Antipodal obstruction). *Let k be even and suppose the symbol F has two roots y and $-y$ with $y \neq 0$. Then $c^2 = a\alpha - \beta = 0$, so the matrix is not admissible in the symmetric class and not admissible in the combinatorially symmetric class either. Hence for even k a period-one certificate requires a symbol whose root set contains no antipodal pair. For odd k no such obstruction exists: there*

$$c^2 = \frac{L_k(y)(y^2 - a^2)}{y},$$

which is not identically zero.

Proof. From $L_k(z+z^{-1}) = z^k + z^{-k}$ we get $L_k(-s) = (-1)^k L_k(s)$, since $s \mapsto -s$ corresponds to $z \mapsto -z$. Let k be even, so $L_k(-y) = L_k(y)$. Writing out $F(y) = 0$ and $F(-y) = 0$ using $F(s) = (s+a)L_k(s) + \alpha s + \beta$ gives

$$aL_k(y) + \alpha y + \beta = -yL_k(y), \quad aL_k(y) - \alpha y + \beta = yL_k(y).$$

Adding gives $\beta = -aL_k(y)$; subtracting and dividing by $2y \neq 0$ gives $\alpha = -L_k(y)$. Therefore $c^2 = a\alpha - \beta = -aL_k(y) + aL_k(y) = 0$. Admissibility requires the spoke weights to be nonzero, and their product is c^2 by Proposition 14, which rules out both classes. For odd k , $L_k(-y) = -L_k(y)$, the same two equations become $aL_k(y) + \alpha y + \beta = -yL_k(y)$ and $-aL_k(y) - \alpha y + \beta = -yL_k(y)$, whence $\beta = -yL_k(y)$ and $\alpha = -aL_k(y)/y$, and $c^2 = a\alpha - \beta = L_k(y)(y^2 - a^2)/y$. $\square$

For $k = 2$ the obstruction is the *only* one, because the symbol then has exactly three roots and the $k - 2$ residual conditions of Theorem 28 are vacuous.

Proposition 19 ($k = 2$ in closed form). *For $k = 2$, prescribing three distinct roots s_1, s_2, s_3 determines $a = -e_1$, $\alpha = e_2 + 2$, $\beta = -e_3 - 2e_1$ with e_i the elementary symmetric functions, and*

$$c^2 = e_3 - e_1e_2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3).$$

So $c^2 = 0$ if and only if two of the roots are antipodal, and $c^2 > 0$ if and only if an odd number of the three pairwise sums is negative.

Proof. Expanding $F = (s+a)(s^2-2) + \alpha s + \beta = s^3 + as^2 + (\alpha-2)s + (\beta-2a)$ and comparing with $\prod_i (s-s_i) = s^3 - e_1 s^2 + e_2 s - e_3$ gives the three formulas, and then $a\alpha - \beta = -e_1(e_2+2) + e_3 + 2e_1 = e_3 - e_1 e_2$. The identity $(s_1 + s_2)(s_1 + s_3)(s_2 + s_3) = e_1 e_2 - e_3$ is elementary. The two consequences follow because the three factors are the pairwise sums. $\square$

Write $\Gamma_n^\circ = \Gamma_n \cap (-2, 2)$, with $M = \lfloor (n-1)/2 \rfloor$ elements $s_1 > \dots > s_M$. For odd n , Γ_n° contains no antipodal pair, since $s_i = -s_j$ forces $i+j = n/2$. For even n the antipodal classes number $n/4$ when $4 \mid n$ and $(n-2)/4$ otherwise, so three pairwise non-antipodal values exist exactly when n is odd with $n \geq 7$ or n is even with $n \geq 12$. This settles $k=2$ completely.

Theorem 20 ($k=2$, all n). *Take the three most negative elements of Γ_n° , that is s_M, s_{M-1}, s_{M-2} . Then*

$$s_{M-1} + s_{M-2} = \begin{cases} -4 \cos(4\pi/n) \cos(\pi/n), & n \text{ odd}, \\ -4 \cos(5\pi/n) \cos(\pi/n), & n \text{ even}, \end{cases}$$

which is negative exactly for odd $n \geq 9$ and for even $n \geq 12$. In those cases all three pairwise sums are negative, so $c^2 > 0$ by Proposition 19 and

$$Z(P(n, 2)) = M(P(n, 2)) = 6 \quad (n \geq 9, n \neq 10).$$

For n divisible by 7 the symbol $\Psi_7 = s^3 + s^2 - 2s - 1$ gives $c^2 = -1$: no symmetric certificate, but the integral combinatorially symmetric matrix

$$A = \begin{pmatrix} I + P + P^{-1} & I \\ -I & P^2 + P^{-2} \end{pmatrix}$$

has nullity exactly 6, so $Z(P(n, 2)) = M_{\text{cs}}(P(n, 2)) = 6$ whenever $7 \mid n$. The remaining n are finite in number and exhaustive computation gives $Z(P(5, 2)) = 5$, $Z(P(6, 2)) = 4$, $Z(P(8, 2)) = 5$ and $Z(P(10, 2)) = 6$.

Proof. For odd n , $M = (n-1)/2$, so $2\pi(M-1)/n = \pi - 3\pi/n$ and $2\pi(M-2)/n = \pi - 5\pi/n$, giving $s_{M-1} + s_{M-2} = -2(\cos(3\pi/n) + \cos(5\pi/n)) = -4 \cos(4\pi/n) \cos(\pi/n)$, negative iff $\cos(4\pi/n) > 0$ iff $n > 8$. For even n , $M = n/2 - 1$ and the same computation with $4\pi/n, 6\pi/n$ gives $-4 \cos(5\pi/n) \cos(\pi/n)$, negative iff $n > 10$. Since s is decreasing, $s_M \leq s_{M-2}$, so $s_M + s_{M-1} \leq s_{M-1} + s_{M-2} < 0$ and $s_M + s_{M-2} \leq s_{M-1} + s_{M-2} < 0$: all three pairwise sums are negative, their product is negative, and $c^2 > 0$. Proposition 14 then gives $\text{null } A = 6$, and $6 \leq M(P(n, 2)) \leq Z(P(n, 2)) \leq 6$ by the rotation bootstrap. The Ψ_7 claim is Theorem 28 with c^2 negative, read in the combinatorially symmetric class where the two spoke weights need only have the prescribed product. $\square$

Remark 21 (What the two exceptions mean). At $n=8$ and $n=10$ the grid is too small to avoid antipodal pairs – $\Gamma_8^\circ = \{\sqrt{2}, 0, -\sqrt{2}\}$ and Γ_{10}° has four elements in two antipodal classes, so by pigeonhole every triple contains such a pair. The two cases then behave differently: $Z(P(8, 2)) = 5$, so there the method is right to fail, while $Z(P(10, 2)) = 6$, so there it is the method that falls short and not the bound. This is the sharpest available statement of the limitation, and it is a statement about Γ_n alone.

Corollary 22 (Excluded cyclotomic symbols, even k). *Let k be even. The roots of Ψ_{2d} are the negatives of the roots of Ψ_d for odd d , and the root set of Ψ_d is closed under negation whenever $4 \mid d$ and $\varphi(d) \geq 4$. Hence in Theorem 28 the factor set D may contain neither both d and $2d$ for any odd d , nor any d divisible by 4 with $\varphi(d) \geq 4$.*

Corollary 22 in fact accounts for *all* of the rejections in the exhaustive enumeration of Proposition 36. At $k = 2$ that enumeration found 17 factor sets in the three-parameter family, of which 5 had $c^2 > 0$ and 12 were rejected. Those 12 split exactly:

$$\underbrace{(3, 8), (3, 12), (4, 8), (4, 12), (6, 8), (6, 12), (3, 4, 6)}_{c^2=0, \text{ every one antipodal}} \quad \underbrace{(7), (9), (4, 5), (5, 6), (6, 10)}_{c^2=-1, \text{ none antipodal}}.$$

The seven on the left are dead in both matrix classes, and Theorem 18 identifies them without computing c^2 at all. The five on the right carry $c^2 = -1$, so each of them *is* realisable in the combinatorially symmetric class. The admissible count at $k = 2$ is therefore 5 symmetrically and 10 in the larger class. The extra symbols cover every n divisible by 7 or by 9; of these only $n = 7$ is out of reach of Theorem 20, and it is precisely the case the larger class was needed for. The antipodal obstruction is the only thing standing in the way of any rational period-one symbol for $k = 2$. Every prediction of Theorem 18 and Corollary 22 was checked against 50 candidate factor sets at $k = 2, 4$ with no violation, and Proposition 19 was verified by exact arithmetic in $\mathbb{Z}[x]/\Phi_n(x)$ over all $\binom{M}{3}$ triples for every $7 \leq n \leq 60$ — 11,000 triples, with $c^2 = 0$ and “contains an antipodal pair” agreeing in every one. See `verification/antipodal_obstruction.py`.

The larger matrix class, and $k = 3$

Theorem 16 separates $c^2 = 0$ from $c^2 < 0$, and that separation matters: only the first is fatal. A symbol with $c^2 < 0$ admits no symmetric certificate, but in the combinatorially symmetric class the two spoke weights are independent and need only have product c^2 , so it is admissible there. Redoing the exhaustive enumeration of Proposition 36 with the weaker condition $c^2 \neq 0$ therefore changes the answer. The case $k = 3$ changes the most.

Theorem 23 ($k = 3$). *Let $F = \Psi_5 \Psi_{10} = (s^2 + s - 1)(s^2 - s - 1) = s^4 - 3s^2 + 1$. Then $F = sL_3(s) + 1$, so $a = \alpha = 0$, $\beta = 1$ and $c^2 = -1$. Hence the integer matrix*

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^3 + P^{-3} \end{pmatrix}$$

has zero diagonal, carries exactly the pattern of $P(n, 3)$, and has nullity exactly $8 = 2k + 2$ for every n divisible by 10. Therefore

$$Z(P(n, 3)) = M_{\text{cs}}(P(n, 3)) = 8 \quad (10 \mid n).$$

Proof. $\Psi_5 = s^2 + s - 1$ and $\Psi_{10} = s^2 - s - 1$, whose product is $s^4 - 3s^2 + 1 = s(s^3 - 3s) + 1 = sL_3(s) + 1$. Matching against $(s + a)L_3 + \alpha s + \beta$ gives $a = 0$, $\alpha = 0$, $\beta = 1$, so $c^2 = a\alpha - \beta = -1$. By Theorem 16 the symbol is non-degenerate, and taking the two spoke weights to be 1 and -1 realises $c^2 = -1$ with both nonzero, so the zero–nonzero pattern is exactly that of $P(n, 3)$. Every root of F is a root of Ψ_5 or Ψ_{10} , hence lies in Γ_n whenever $10 \mid n$, and all four roots are interior, so each contributes 2: null $A = 8$. The upper bound $Z(P(n, 3)) \leq 8$ is the rotation bootstrap, and $M_{\text{cs}} \leq Z$ holds because the forcing argument uses only the zero–nonzero pattern. $\square$

Remark 24 (Two things this certificate illustrates). First, A is the adjacency matrix of $P(n, 3)$ with one set of spokes negated, and the sign matters entirely: the unsigned adjacency matrix has symbol $sL_3 - 1$ and nullity 0 for every n divisible by 10, consistent with Theorem 29. Second, the four roots of F are $\pm\varphi$ and $\pm\varphi^{-1}$ — *two antipodal pairs*. By Corollary 18 such a symbol would be degenerate for every even k ; because 3 is odd it is not. So the parity corollary does not only rule things out, it says where to look.

The same enumeration in the larger class yields five further symbols at $k = 2$ (covering n divisible by 7, 9, 20 or 30), five more at $k = 3$ beyond Theorem 23 (n divisible by 24, 40 or 42), three at $k = 4$ (60, 70, 90) and two at $k = 5$ (126); all have $c^2 \in \{-1, -2\}$. At $k = 6$ it yields nothing, and now for a reason.

The signed adjacency matrix, and $k = 7$

Both the $k = 3$ certificate and one of the $k = 7$ symbols found by this enumeration have $a = \alpha = 0$ and $\beta = 1$, that is, symbol $F = sL_k(s) + 1$. That is not a coincidence, and identifying what it means gives an exact value at $k = 7$.

With $a = \alpha = 0$ both diagonals vanish, so A is the adjacency matrix of $P(n, k)$ with one set of spokes negated. Substituting $s = z + z^{-1}$, $L_k = z^k + z^{-k}$ and clearing denominators turns $F(s) = 0$ into

$$Q_k(z) := z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1 = 0,$$

which is exactly the adjacency polynomial $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$ of Theorem 29 with the sign of the middle term reversed — the spoke sign flip, visible in the symbol.

Theorem 25 (Signed adjacency classification). *Let A^- be the adjacency matrix of $P(n, k)$ with one set of spokes negated. Then*

$$\text{null } A^- = \sum \varphi(d) \quad \text{over those } d \text{ with } \Phi_d \mid Q_k \text{ and } d \mid n,$$

by the argument of Theorem 29 applied to Q_k in place of P_k . In particular $\text{null } A^- = 2k + 2$ for suitable n exactly when Q_k is a product of cyclotomic polynomials.

So the question of which k admit an *integer* certificate of this shape becomes: for which k does Q_k split into cyclotomics? This is decidable for each k , and the answer is rigid. For $2 \leq k \leq 60$ it happens only for

$$k = 2 : Q_2 = \Phi_5\Phi_6, \quad k = 3 : Q_3 = \Phi_5\Phi_{10}, \quad k = 7 : Q_7 = \Phi_5\Phi_{10}\Phi_{24},$$

with $\sum \varphi(d)$ equal to 6, 8, 16 — the ceiling $2k + 2$ in every case — and lcm equal to 30, 10, 120. The unsigned P_k splits only for $k = 2$ and $k = 5$ (Theorem 29), so between them the two sign choices give integer adjacency-type certificates exactly at $k \in \{2, 3, 5, 7\}$.

Theorem 26 ($k = 7$). $Q_7 = \Phi_5\Phi_{10}\Phi_{24}$, so for every n divisible by 120 the integer matrix

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^7 + P^{-7} \end{pmatrix}$$

carries the pattern of $P(n, 7)$ and has nullity exactly $16 = 2k + 2$. Hence

$$Z(P(n, 7)) = M_{\text{cs}}(P(n, 7)) = 16 \quad (120 \mid n).$$

This replaces a numerical claim by an exact one. Section 10 reports $Z(P(96, 7)) = 16$ via a period-4 certificate verified in floating point; Theorem 26 gives the same value for infinitely many n from a matrix of zeros and ± 1 , with the nullity computed by a cyclotomic factorisation rather than by an eigenvalue. Confirmed numerically: the matrix has nullity 16 at $n = 120$, and 6, 8 at the corresponding n for $k = 2, 3$ (`verification/signed_adjacency.py`).

Theorem 27 ($k = 6$ is impossible over $\mathbb{Q}$). *Of the 225 products $\prod_{d \in D} \Psi_d$ of total degree 7 with every Ψ_d of interior roots, 216 do not lie in the three-parameter family, and all 9 that do are degenerate in the sense of Theorem 16: each is $(s + a)(L_6 + t)$ with $a \in \{0, \pm 1\}$ and $t \in \{0, \pm 1\}$, namely Ψ_d -products over $D = (3, 8, 24), (4, 8, 24), (6, 8, 24), (3, 9, 18), (4, 9, 18), (6, 9, 18), (3, 36), (4, 36), (6, 36)$. Hence no rational period-one symbol attains 14 for $k = 6$, in either matrix class.*

This is stronger than the statement it replaces. Where we previously reported that a search found nothing at $k = 6$, the obstruction now identifies every candidate that survives the family condition and shows each one is degenerate, and Corollary 17 bounds the degenerate list *a priori* at 25 polynomials. The $k \geq 6$ certificates of Section 10 therefore have to break the symmetry, and they do.

For $d \geq 3$ let $\Psi_d \in \mathbb{Z}[s]$ be the minimal polynomial of $2 \cos(2\pi/d)$, of degree $\varphi(d)/2$, whose roots are the primitive values $2 \cos(2\pi j/d)$ with $\gcd(j, d) = 1$; put $\Psi_1 = s - 2$, $\Psi_2 = s + 2$. Every root of Ψ_d lies in Γ_n as soon as $d \mid n$.

Theorem 28 (Cyclotomic certificates). *Let D be a set of distinct positive integers with $\sum_{d \in D} \deg \Psi_d = k + 1$ and put $F = \prod_{d \in D} \Psi_d$. Write $G = F - sL_k$, let a be the coefficient of s^k in G , and suppose $G - aL_k$ has degree at most 1, say $G - aL_k = \alpha s + \beta$; that is $k - 2$ linear conditions with integer coefficients. Put $e = 1$, $d = \alpha$, $c^2 = a\alpha - \beta$, and suppose $c^2 > 0$. Then for every n divisible by $\text{lcm}(D)$ with $2k < n$, the matrix*

$$A = \begin{pmatrix} aI + P + P^{-1} & cI \\ cI & \alpha I + P^k + P^{-k} \end{pmatrix}$$

lies in $S(P(n, k))$ and has nullity $2 \sum_{d \in D, d \geq 3} \deg \Psi_d + \#\{d \in D : d \leq 2\}$. If every $d \in D$ is at least 3 this equals $2k + 2$, and then

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2.$$

Proof. By construction $F = (s + a)L_k + \alpha s + \beta$ as an identity in $\mathbb{Z}[s]$, so F is the symbol of A by Proposition 14, and $A \in S(P(n, k))$ because $c^2 > 0$ makes c real and nonzero while the edge weights 1 are nonzero. Each $d \in D$ divides n , so all $k + 1$ roots of F lie in Γ_n ; they are distinct because the Ψ_d are distinct irreducible polynomials. Counting multiplicities as in Proposition 14 gives the stated nullity, and the last line combines it with (1) and Theorem 3. $\square$

A search over $k \leq 12$ and $d \leq 210$ (`verification/cyclotomic_certificates.py`) gives the following. Every certificate listed has $a, \alpha, c \in \mathbb{Z}$, hence is an *integer* matrix, and its nullity was confirmed by exact rational Gaussian elimination.

Two of these deserve to be written out. For $k = 2$ and $k = 5$ the parameters are $a = d = 0$, $c = 1$: the certificate is the *adjacency matrix* of $P(n, k)$, and the symbol is $sL_k(s) - 1$, with

$$sL_2 - 1 = (s + 1)(s^2 - s - 1) = \Psi_3 \Psi_{10}, \quad sL_5 - 1 = (s - 1)(s + 1)(s^4 - 4s^2 + 1) = \Psi_6 \Psi_3 \Psi_{24}.$$

Equivalently, since $\det M_m = s_m t_m - 1$,

$$\text{null}(\text{Adj } P(n, k)) = \#\left\{m \in \mathbb{Z}_n : 2 \cos \frac{2\pi(k+1)m}{n} + 2 \cos \frac{2\pi(k-1)m}{n} = 1\right\},$$

a rational linear relation among cosines of rational angles — the setting of the Conway–Jones classification. That question has a complete answer, given in Theorem 29 below. (An earlier version of this paper asserted that “in the computed range $k \leq 10$ only $k = 2, 5, 8$ ” admit solutions. That was drawn from a scan filtered at nullity ≥ 6 and is wrong: $k = 4, 7, 10, 11, 12$ also admit

k	D	$\text{lcm}(D)$	F	a	d	c^2	nullity
2	$\{3, 10\}$	30	$sL_2 - 1$	0	0	1	$6 = 2k + 2$
2	$\{14\}$	14	Ψ_{14}	-1	0	1	$6 = 2k + 2$
3	$\{1, 4, 5\}$	20	$\Psi_1 \Psi_4 \Psi_5$	-1	-1	1	7
4	$\{4, 30\}$	60	$(s+1)L_4 - s - 2$	1	-1	1	$10 = 2k + 2$
4	$\{5, 14\}$	70	$\Psi_5 \Psi_{14}$	0	1	1	$10 = 2k + 2$
4	$\{5, 18\}$	90	$\Psi_5 \Psi_{18}$	1	0	1	$10 = 2k + 2$
5	$\{3, 6, 24\}$	24	$sL_5 - 1$	0	0	1	$12 = 2k + 2$

Table 2: Exact period-one certificates. The $k = 2$, $k = 4$ and $k = 5$ rows give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every n divisible by the stated modulus.

solutions, with smaller nullity.) For $k = 5$ the count is $12 = 2k + 2$ whenever $24 \mid n$: the plain adjacency matrix of $P(n, 5)$ has nullity 12, which settles Z for that family. For $k = 4$ the symbol is $(s+1)L_4(s) - s - 2 = s(s^4 + s^3 - 4s^2 - 4s + 1) = \Psi_4 \Psi_{30}$, giving the integer certificate with $a = 1$, $d = -1$, $c = 1$.

The adjacency nullity, classified completely

Since $\det M_m = s_m t_m - 1$, the adjacency matrix is singular at m exactly when $z = \omega^m$ satisfies $z^{k+1} + z^{-(k+1)} + z^{k-1} + z^{-(k-1)} = 1$. Clearing denominators gives an integer polynomial, and the question becomes which of its roots are roots of unity.

Theorem 29. Put $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1 \in \mathbb{Z}[z]$. Then

$$\text{null}(\text{Adj } P(n, k)) = \sum_{\substack{d: \Phi_d \mid P_k \\ d \mid n}} \varphi(d),$$

the sum over those d whose cyclotomic polynomial divides P_k . The list of such d is finite and computable once per k .

Proof. m contributes iff ω^m is a common root of P_k and $z^n - 1$, i.e. a root of $\gcd(P_k, z^n - 1)$. The roots of unity among the roots of P_k are exactly the roots of its cyclotomic factors Φ_d , and Φ_d 's roots are n -th roots of unity precisely when $d \mid n$; each such Φ_d contributes its $\varphi(d)$ roots. Finiteness is immediate from $\deg P_k = 2k + 2$, and is the elementary shadow of the Conway–Jones theorem on rational linear relations among cosines of rational angles. $\square$

k	$2k + 2$	cyclotomic factors of P_k	lcm	nullity when $\text{lcm} \mid n$
2	6	Φ_3, Φ_{10}	30	$6 = 2k + 2$
3	8	none	—	0 for every n
4	10	Φ_3	3	2
5	12	$\Phi_3, \Phi_6, \Phi_{24}$	24	$12 = 2k + 2$
6	14	none	—	0 for every n
7	16	Φ_3, Φ_6	6	4
8	18	Φ_3, Φ_{10}	30	6

So the adjacency matrix of $P(n, k)$ is *never* singular for $k = 3, 6, 9$, whatever n ; and it attains the full ceiling $2k+2$ only for $k = 2$ and $k = 5$. The formula was checked against a direct count for $k = 2, 5, 8$ at every $13 \leq n < 200$, with no discrepancy (`verification/adjacency_classification.py`).

9 The general setting: cyclic covers

Nothing in Section 8 used the particular shape of $P(n, k)$ beyond the presence of the rotation. It is worth saying what the natural setting is.

Let B be a finite graph whose edges carry *voltages* in $\mathbb{Z}_n$ (no edge of voltage 0 at a loop, since that would lift to a self-loop). The *derived graph* B^n has vertex set $V(B) \times \mathbb{Z}_n$, with (x, i) joined to $(y, i + v)$ for each edge $x - y$ of voltage v . This is the standard cyclic cover of topological graph theory. Taking B to be two vertices u, v with a loop of voltage 1 at u , a loop of voltage k at v , and an edge $u - v$ of voltage 0 recovers $B^n = P(n, k)$ exactly.

Theorem 30 (Cover reduction). *Let A carry the pattern of B^n and commute with the $\mathbb{Z}_n$ action. Then A block-diagonalises into n blocks of size $|V(B)|$,*

$$M_m[x, y] = \sum_{\text{edges } x \rightarrow y \text{ of voltage } v} w_{xy} \omega^{mv} + \delta_{xy} a_x,$$

and $\text{null } A = \sum_m \text{null } M_m$. Writing $\det M(\zeta)$ as a Laurent polynomial in ζ , the number of m at which a block is singular — hence $\text{null } A$ — is at most the degree span of $\det M(\zeta)$, which is determined by the voltages of B alone.

For $P(n, k)$ the degree span is $2k + 2$, recovering Corollary 12 in the equivariant case. The reduction and the ceiling were verified against independently constructed full matrices, with weights tuned so that the nullity is genuinely nonzero, on seven base graphs on two to five vertices with assorted voltages (`verification/cyclic_covers.py`).

Two caveats. Theorem 30 is stated for *equivariant* A ; the stronger statement of Theorem 11, which bounds the nullity of *every* matrix carrying the pattern, we have proved only for $P(n, k)$, where the nonzero spoke weights permit the elimination. Whether the corresponding elimination can be carried out over an arbitrary base graph — and so whether the ceiling is a property of the cover rather than of equivariant matrices on it — we leave open. Second, we have tested the non-equivariant bound on general covers by *maximising* the nullity rather than sampling — random matrices are generically nonsingular, so sampling proves nothing. Optimising over fully non-equivariant weights on four small covers, the largest attainable nullity never exceeded the degree span, and equalled it for the theta graphs with voltages $(0, 1, 2)$ and $(0, 1, 3)$; on the $P(n, 2)$ and $P(n, 3)$ covers at $n = 9$ it fell short, as it must, since $Z(P(9, k)) < 2k + 2$ there. That is evidence for the conjecture, not a proof of it.

The construction on a general cover

Section 8 prescribes a symbol for $P(n, k)$ and solves for the weights. Nothing in that used the base graph either. For any cover, writing $\det M(\zeta)$ in $\sigma = \zeta + \zeta^{-1}$ gives a polynomial of degree $D/2$ whose roots in Γ_n are the singular blocks, so the same move applies: prescribe $D/2$ interior grid values, solve for the weights, and the nullity is D .

Theorem 31 (Attainment on a cover). *Let B^n be a cyclic cover and suppose the prescribed-symbol construction yields A carrying the pattern of B^n with $\text{null } A = \nu$. Then*

$$Z(B^n) \geq \nu,$$

by $M \leq Z$. In particular $Z(B^n) \geq D$ whenever the construction attains the ceiling.

This is a statement about all cyclic covers, with $P(n, k)$ — base graph two loops of voltages 1 and k and one edge of voltage 0, $D = 2k + 2$ — as the special case. We have verified attainment of the ceiling on four different base graphs: the $P(n, 2)$ base; the theta graph with voltages $(0, 1, 3)$, a different cubic family; a three-vertex path base with $D = 12$, whose cover has $3n$ vertices; and a four-vertex star base with $D = 12$. In each the constructed matrix has nullity exactly D with every required entry a substantial fraction of the matrix scale.

We also tested the ceiling sharply, which the earlier check did not. Asking whether $\ker A$ injects into a window of D consecutive fibre coordinates is vacuous when the nullity is 2 and D is 12; it bites only at the ceiling. Repeating it on certificates of nullity exactly D , injectivity holds in every case — including the star base, which has *no* Hamiltonian path, so the ordering hypothesis under which the $P(n, k)$ elimination generalises is sufficient but not necessary.

Remark 32 (The hypothesis is needed). Theorem 31 is conditional on the construction succeeding, and that condition is not vacuous. For the $P(n, 2)$ base, $D = 6$, the construction attains 6 at $n = 7, 9, 11, 12$ but fails at $n = 8$ and $n = 10$; and $Z(P(8, 2)) = 5 < 6$ by exhaustive search, so the unconditional statement $Z(B^n) \geq D$ is *false*. What does hold unconditionally is $Z(B^n) \geq \nu$ for the attained ν , which we checked against exhaustively computed zero forcing numbers on covers with 14 to 20 vertices, with no violation. Whether the construction succeeds is not a dimensional question: $n = 7$ attains the ceiling with no slack in the choice of roots while $n = 10$ fails with slack to spare. For the $P(n, 2)$ base the criterion is now known exactly — it is Proposition 19, and $n = 8, 10$ fail because every triple on those grids contains an antipodal pair. For a general base we do not have the analogue.

Remark 33 (Where this sits). The reduction is a discrete Floquet theory. The monodromy of Theorem 11 plays the role of the Hill discriminant for a periodic Jacobi operator, and the condition $T = I$ — all Floquet multipliers roots of unity — is the statement that the operator has purely periodic spectrum at the relevant energy. The parameter $M(G)$ itself belongs to the minimum rank programme for graphs, and $Z(G)$ arose independently in the controllability of quantum spin networks and in sensor placement for power grids.

Certificates beyond the rational ones

Theorem 28 is sufficient but not necessary: F need not have rational coefficients, only real ones, and its roots need only lie in Γ_n , not form full Galois orbits. By (2) every period-one symbol with three or more grid roots arises from some triple of them, so running over all triples of Γ_n finds the true period-one optimum for each (n, k) . Candidates so found are re-checked in exact arithmetic in $\mathbb{Q}(\zeta_n)$, representing $2 \cos(2\pi m/n)$ as $x^m + x^{n-m}$ modulo $\Phi_n(x)$: the membership identity, the reality of a, α, β, c^2 (they are fixed by $x \mapsto x^{n-1}$) and $c^2 \neq 0$ are all decided exactly, and the sign of the nonzero algebraic number c^2 is then read off at high precision. This is not pedantry — see the remark on numerical traps below.

Theorem 34. *Each of the following holds, with an explicit period-one certificate verified exactly in $\mathbb{Q}(\zeta_n)$:*

$$\begin{aligned} Z(P(n, 3)) &= M(P(n, 3)) = 8 && \text{for } n = 40, 60, 80, 120, \\ Z(P(n, 4)) &= M(P(n, 4)) = 10 && \text{for } n = 60, 70, 90, 120, \\ Z(P(n, 5)) &= M(P(n, 5)) = 12 && \text{for } n = 24, 48, 72, 96, 120, \\ Z(P(n, 2)) &= M(P(n, 2)) = 6 && \text{for } 9 \leq n \leq 130, n \neq 10. \end{aligned}$$

The $k = 4$ and $k = 5$ lines are the first exact values of $Z(P(n, k))$ for $k \geq 4$, the range [2] records as open; previously we had only $8 \leq Z(P(n, 4)) \leq 10$ and nothing at all for $k = 5$. The $k = 3$

line is a matrix proof of values [2] obtains combinatorially. Some certificates are irrational: at $(n, k) = (40, 3)$ one has $a = -\sqrt{2}$, $d = 0$, $c^2 = 1$, and at $(60, 3)$, $c^2 = 1/\varphi$ with φ the golden ratio. Full output is in `results/zero_forcing/certified_lower_bounds.txt`.

Remark 35 (Why exact arithmetic, twice over). Counting grid roots by $|F(r)| < \varepsilon$ is not safe. With $\varepsilon = 10^{-8}$ the scan reported nullity 7 for $k = 2$ at $n = 126, 128, 130$ — impossible, since $2k + 2 = 6$ — because Γ_n becomes dense and near-roots pass the test. Separately, a certificate at $(n, k) = (48, 7)$ with $|a|$ enormous against the edge weights made a threshold relative to $\max |A|$ report nullity 48. Both were caught by Corollary 12, which bounds the nullity by $2k + 2$ and so serves as an automatic correctness check on any claimed certificate, exactly as the inequality $Z \geq M$ did for the earlier period-two search. The fixes are to test each Fourier block against its own scale, and to certify in $\mathbb{Q}(\zeta_n)$ rather than in floating point.

Where the method stops

Running the exhaustive triple scan for $2 \leq k \leq 8$ and $n \leq 130$ gives the period-one optimum exactly (`verification/period1_optimum.py`):

k	$2k + 2$	best period-one nullity	attained at
2	6	6	every n with $9 \leq n \leq 130$ except $n = 7, 8, 10$
3	8	8	$n = 40, 60, 80, 120$
4	10	10	$n = 60, 70, 90, 120$
5	12	12	$n = 24, 48, 72, 96, 120$
6	14	8	$n = 35, 37, 70, 74, 105, 111$
7	16	10	$n = 60, 120$
8	18	8	$n = 21, 26, 27, 39, \dots$

Table 3: The period-one construction reaches the ceiling $2k + 2$ for $k \leq 5$ and, in the computed range, falls well short for $k \geq 6$.

Proposition 36 (The rational search is complete). *Fix k . A period-one symbol attaining $2k + 2$ must be squarefree with all $k + 1$ roots interior and on Γ_n ; if in addition it has rational coefficients it is a product of distinct Ψ_d with $\sum \deg \Psi_d = k + 1$. Every factor then satisfies $\deg \Psi_d \leq k + 1$, i.e. $\varphi(d) \leq 2k + 2$, which admits only finitely many d , all listable. Hence for each k the search over rational symbols is a finite complete enumeration. Carrying it out (`verification/cyclotomic_exhaustive.py`):*

k	$2k + 2$	candidates	rejected		admissible	
			outside family	$c^2 = 0$	sym	cs
2	6	17	0	7	5	5 more
3	8	35	21	8	0	6
4	10	67	54	7	3	3 more
5	12	127	118	6	1	2 more
6	14	225	216	9	0	0
7	16	385	378	6	0	1
8	18	651	644	7	0	0
9	20	1065	1057	8	0	0
10	22	1705	1698	7	0	0

The last two columns are the point. The original enumeration reported only the “sym” column and recorded $k = 3$ and $k = 7$ as impossible for rational symbols; they are not. By Theorem 16 only the $c^2 = 0$ column is fatal, and every remaining rejection has $c^2 \in \{-1, -2\}$, hence is realisable in the combinatorially symmetric class. This is what yields Theorem 23 at $k = 3$ and Theorem 26 at $k = 7$. Note also that the $c^2 = 0$ counts are 7, 8, 7, 6, 9, 6, 7, 8, 7, never more than nine, exactly as Corollary 17 predicts from its bound of 25.

Two further things follow. First, $k = 6, 8, 9$ and 10 really are impossible over $\mathbb{Q}$ in both classes, and now for an identified reason rather than a failed search: every candidate that clears the family condition is one of the finitely many degenerate symbols. Second, the rejections are overwhelmingly of one kind — at $k = 7$, 378 of the 385 candidates fail because F is *not in the three-parameter family*. The binding constraint is therefore the $k - 2$ linear membership conditions imposed on a 3-dimensional family, and it tightens as k grows. That identifies the fix: enlarge the family.

How far must one enlarge it? A period- d matrix has $5d$ weights, but most of that freedom is gauge: conjugating by a positive diagonal matrix preserves both the pattern and the nullity. What matters is the dimension of the set of symbols actually reachable. Writing G for the degree- $(k + 1)$ polynomial in $\sigma = \zeta + \zeta^{-1}$ obtained from $\det M_\ell$, we computed the rank of the Jacobian of the map (weights) $\mapsto$ (monic coefficients of G) at random points, for $3 \leq k \leq 13$ and $1 \leq d \leq 7$ (`verification/period_d_rank.py`). In every case

$$\text{rank} = \min(2d + 1, k + 1).$$

So the reachable symbols form a $(2d + 1)$ -dimensional family — not the $5d - 1$ -dimensional one a naive parameter count suggests — and the map is a submersion, so that every G is locally reachable, precisely when

$$2d + 1 \geq k + 1, \quad \text{that is} \quad d \geq \lceil k/2 \rceil.$$

At $d = 1$ this returns the 3 of Proposition 14, which is a check on the computation. We record the law as an observation supported by computation, not as a theorem: we have not proved the rank formula, only verified it over that range. Its consequence, if it holds in general, is that the period- d construction should reach $2k + 2$ for every k once $d \geq \lceil k/2 \rceil$ and n/d is large enough to host $k + 1$ grid roots — a construction whose modulus is d rather than an lcm of cyclotomic orders, and therefore one that would cover a fixed congruence class of n for every k .

So the construction settles $k \leq 5$ on those residue classes and says nothing useful for $k \geq 6$; whether that is a limitation of period one or of $n \leq 130$ we do not know. For n outside the classes above, no certificate of nullity $2k + 2$ was found at all. This is not for want of parameters. Dropping equivariance

entirely leaves $5n$ free weights in the symmetric class and $8n$ in the combinatorially symmetric one, whereas a nullity of $r = 2k + 2$ costs $r(r + 1)/2 = (k + 1)(2k + 3)$ conditions in the first case and r^2 in the second — budgets that are met comfortably at every graph we tried. A two-stage search (many random starts on $\sum_{j \leq r} \lambda_j^2$ with analytic gradients, then a Gauss–Newton polish of the bilinear system $A(w)X = 0$, $X^\top X = I$; `verification/fast_nullity.py`) attains nullity 6 at machine precision on $P(12, 2)$ and $P(14, 2)$ but plateaus near 10^{-4} for $r = 7, 8$ on $P(12, 3)$ over hundreds of independent starts, in both matrix classes. We therefore record, as an observation rather than a theorem, that the obstruction to $2k + 2$ at general n appears to be arithmetic — whether Γ_n contains a compatible configuration of $k + 1$ points — and not a matter of dimension. Two guardrails are worth stating, since we tripped on both: the combinatorially symmetric class *contains* the symmetric one, so any search reporting a smaller answer for it is faulty (ours did, on $P(12, 2)$, until the larger search was seeded with the symmetric certificate); and any claimed nullity above $2k + 2$ is a bug by Corollary 12.

Why every result so far lives on a congruence class

Each exact value in this section holds for n in a divisibility class, not for all large n . That is not an artefact of our search. It is forced, and the following says exactly how.

Theorem 37 (Period-one certificates are bound to congruence classes). *Fix $k \geq 1$ and a period-one matrix A with parameters (a, α, β) and symbol F .*

- (i) *Suppose $\text{null } A = 2k + 2$ for some $n > 2k$. Then every root of F is $2\cos(2\pi j/d)$ with $\gcd(j, d) = 1$ and $d \geq 3$, and such a root lies in Γ_m if and only if $d \mid m$. Hence the set of n at which this same A attains the ceiling is exactly $\{n : \text{lcm}(D) \mid n\}$, where D is the multiset of those d . It is therefore a single divisibility class of density $1/\text{lcm}(D) \leq 1/3$, and it contains at most one prime.*
- (ii) *Consequently, to attain the ceiling for infinitely many n outside one divisibility class — in particular for infinitely many primes, or for all large n — the weights must depend on n .*
- (iii) *For $k \geq 3$ that dependence is constrained by one universal identity: the coefficient of s^{k-1} in F equals $-k$ for every (a, α, β) , so the roots always satisfy*

$$e_2(r_1, \dots, r_{k+1}) = -k, \quad \text{equivalently} \quad \sum_i r_i^2 = e_1^2 + 2k.$$

For $k = 2$ the coefficient of $s^{k-1} = s$ is $\alpha - 2$, which is free, so no such condition exists.

Proof. (i) Attaining $2k+2$ needs all $k+1$ roots interior and in Γ_n (Proposition 14), and an element of $\Gamma_n \cap (-2, 2)$ is $2\cos(2\pi m/n)$, which written in lowest terms is $2\cos(2\pi j/d)$ with $d \mid n$, $d \geq 3$. Conversely $2\cos(2\pi j/d) \in \Gamma_m$ iff $d \mid m$. The roots of F do not move with n , since F depends only on (a, α, β) ; so all of them lie on Γ_n exactly when every d divides n . A set $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only if L is itself prime, and then only L .

(ii) Immediate from (i): a fixed A covers one class, and a union of finitely many classes still omits all but finitely many primes.

(iii) Write $(s+a)L_k = sL_k + aL_k$. Since L_k has nonzero coefficients only in degrees congruent to k modulo 2, aL_k has no s^{k-1} term, and the s^{k-1} coefficient of sL_k is the s^{k-2} coefficient of L_k , namely $-k$. The terms αs and β reach degree $k-1$ only when $k \leq 2$. For a monic polynomial of degree $k+1$ the coefficient of s^{k-1} is e_2 , and $\sum r_i^2 = e_1^2 - 2e_2$. $\square$

Part (iii) is what separates $k = 2$ from everything else. With no residual condition, Theorem 20 is free to choose the roots *as functions of n* — the three most negative elements of Γ_n° — and that is precisely why $k = 2$ is general in n while no other k is. For $k \geq 3$ an n -dependent choice must satisfy $e_2 = -k$ identically along the family. Substituting $r_i = 2\cos(2\pi m_i/n)$ and expanding e_1^2 by the product-to-sum formula turns this into a rational linear relation among cosines of rational angles set equal to a rational number. Relations of that shape are the subject of Conway–Jones [3], and they are arithmetically rigid: they hold not on open families but on sparse, classifiable configurations. That rigidity is the mechanism behind every divisibility condition in this paper, and it is the same mechanism as in Theorem 29.

We verified the identity of (iii) against all 26 certificates found in this paper: $e_2 = -k$ in every case with $k \geq 3$, and e_2 takes six different values among the $k = 2$ certificates, as (iii) predicts (`verification/general_n.py`).

Remark 38 (What this licenses, and what it forecloses). Theorem 37 should be read as a demarcation rather than a limitation. It says the entire cyclotomic layer of this paper — every result in Sections 8 and 10 — is *provably* incapable of producing a statement valid for all large n , so the congruence conditions are a boundary of the method and not a shortfall in its execution. It also says where to go instead: Corollary 12 characterises the ceiling for *every* matrix on the pattern, not only the invariant ones, by the condition $T = I$ on the monodromy, and that condition carries no arithmetic in it.

10 Period- d certificates, and the first values for $k \geq 6$

Proposition 36 leaves $k = 6$ and $k \geq 8$ untouched: no rational period-one symbol attains $2k + 2$ there in either matrix class, and the exhaustive triple scan of Section 8 reaches only 8, 10, 8 for $k = 6, 7, 8$ over $n \leq 130$. ($k = 7$ is settled exactly by Theorem 26; what follows gives it, and $k = 6$ and $k = 8$, for further n .) The rank law says why, and what to do about it.

Prescribing the symbol

For $d \geq \lceil k/2 \rceil$ the map (weights) $\mapsto$ (monic coefficients of G) is a submersion, so instead of *searching* for a certificate one can *prescribe* one: choose $k + 1$ interior values of Γ_m , $m = n/d$, form the monic G^* with exactly those roots, and solve $G(w) = G^*$ for the weights. This is $k + 1$ equations in $5d$ unknowns with a full-rank Jacobian, so the solution set is a manifold of dimension $5d - (k + 1)$ — eight-dimensional for $k = 6$, $d = 3$. Each root is interior, hence met by two Fourier indices ℓ and $m - \ell$, so all $2k + 2$ of the corresponding blocks are singular and the nullity is $2k + 2$ by Corollary 12.

Two conditions must both hold, and they are independent: $m \geq 2k + 4$, so that Γ_m contains $k + 1$ distinct *interior* values at all; and $d \geq \lceil k/2 \rceil$, so that the symbol is reachable. For $k = 6$ this forces $d \geq 3$ and $n \geq 48$.

A degeneracy, and how it is escaped

Solving $G(w) = G^*$ without further constraint reliably returns a point at which one required-nonzero weight — in every case a spoke — has collapsed to zero. Such a point is a genuine solution of $G(w) = G^*$, because with a spoke weight zero the matrix block-diagonalises and G factors as $\det U_\ell \cdot \det V_\ell$; but the matrix then carries the pattern of a *different* graph and certifies nothing. At these points the Jacobian rank drops from 7 to 6: they are singular points of the map, which is why repeated random starts return to them.

Neither a realisability barrier nor continuation escapes them. Driving the vanishing spoke away from zero while re-solving loses the symbol at the first step ($\tau = 0.02$, error 9×10^{-5}). What does escape is *slack in the choice of roots*. At $n = 48$, Γ_{16} has exactly 7 interior values against the 7 required, so the root set is unique and there is nothing to vary; at $n = 54$ there are 8 interior values, giving 8 root sets, and all eight degenerate. At $n = 60$ and $n = 72$ there are 9 and 11 interior values, hence 36 and 330 root sets, and realisable points exist.

Theorem 39.

$$\begin{aligned} Z(P(60, 6)) &= M(P(60, 6)) = 14, & Z(P(72, 6)) &= M(P(72, 6)) = 14, \\ Z(P(96, 7)) &= M(P(96, 7)) = 16, & Z(P(96, 8)) &= M(P(96, 8)) = 18. \end{aligned}$$

In each case the period is the one the rank law prescribes: $d = 3$ for $k = 6$ and $d = 4$ for $k = 7, 8$, i.e. $d = \lceil k/2 \rceil$.

These are the first exact values of $Z(P(n, k))$ for $k \geq 6$. That the required period was predicted in advance, and that $d = 3$ then failed for $k = 7, 8$ while $d = 4$ succeeded, is the strongest evidence we have that the rank law is describing something real rather than fitting the cases it was measured on. The certificates are period-three matrices found by the construction above and verified on the full $2n \times 2n$ matrix, independently of the Fourier blocks used to find them. For $n = 60$ with roots at $\ell \in \{2, 3, 4, 5, 6, 8, 9\}$ the 120×120 matrix has its 14 smallest eigenvalues below 4×10^{-15} and its fifteenth at 5.1×10^{-2} — a separation of thirteen orders of magnitude — every vertex carries exactly three off-diagonal nonzeros, and the smallest required entry is 0.188 of the matrix scale. For $n = 72$ with $\ell \in \{1, 4, 6, 7, 9, 10, 11\}$ the corresponding figures are nullity 14, first nonzero eigenvalue 1.2×10^{-3} of scale, and smallest required entry 0.162. For $k = 7$ at $n = 96$, $d = 4$, $\ell \in \{3, 5, 6, 7, 8, 9, 10, 11\}$: nullity 16, sixteen eigenvalues below 1.4×10^{-14} , smallest required entry 0.107. For $k = 8$ at $n = 96$, $d = 4$, $\ell \in \{2, 3, 4, 5, 6, 7, 8, 9, 11\}$: nullity 18, first nonzero eigenvalue 1.25×10^{-2} of scale, smallest required entry 0.260. Several further root sets at each n give independent certificates — three at $k = 8$ alone, the best with smallest required entry 0.362 — so these are not isolated points but regions.

k	$2k + 2$	period d	$\lceil k/2 \rceil$	n	smallest required entry
6	14	3	3	60, 72	0.188, 0.162
7	16	4	4	96	0.107
8	18	4	4	96	0.260

Table 4: Period- d certificates attaining the ceiling for $k \geq 6$. The period used is in every case the one the rank law prescribes.

Why an exact certificate is hard here

The $k \geq 6$ certificates are numerical because their weights are algebraic numbers of no recognised form. The obvious remedy — find *rational* weights, so that the nullity can be confirmed by exact rational elimination as for $k = 4, 5$ — runs into a structural obstruction.

Proposition 40. *Let A be a period- d matrix on $P(n, k)$ with rational weights, and $m = n/d$. The Fourier blocks have entries in $\mathbb{Q}(\zeta_m)$, and the Galois automorphism $\sigma_j : \zeta_m \mapsto \zeta_m^j$ carries M_ℓ to $M_{j\ell}$. Since σ_j fixes the weights, $\sigma_j(\det M_\ell) = \det M_{j\ell}$, so*

$$\det M_\ell = 0 \implies \det M_{j\ell} = 0 \quad \text{for every } j \text{ coprime to } m.$$

*Hence the set of singular blocks is a union of full Galois orbits of Γ_m , and a rational certificate can only realise a **Galois-closed** root set.*

This is checked numerically in the form it can be: for rational weights the product of $\det M_\ell$ over a Galois orbit — the norm — comes out exactly rational, to machine zero, at $(k, d, n) = (6, 3, 60)$, $(6, 3, 72)$, $(6, 4, 64)$, $(5, 1, 24)$ and $(7, 4, 96)$.

The obstruction is now visible: exactness requires a Galois-closed root set, and Galois-closed root sets are precisely the ones that degenerate. All eight at $(k, d, n) = (6, 3, 72)$ hit the symbol to 10^{-15} with nullity 14 and all eight collapse a weight; both at $(6, 3, 60)$ likewise. So for these parameters *no rational certificate exists*.

Relaxing to a subfield does not obviously help, and it is worth being precise about why. If the weights lie in $K = \mathbb{Q}(\zeta_m)^H$ then the singular set need only be H -stable, and at $(6, 3, 72)$ all three

index-two subgroups admit realisable root sets. But that implication runs one way: an H -stable root set being realisable shows only that the necessary condition is not violated, and gives no reason to expect a certificate for it to have weights in K . Indeed integer-relation detection finds no such representation for the certificates we have, a search parametrised inside $\mathbb{Q}(\sqrt{2})$ from the start returns nothing, and at $(6, 3, 60)$ every index-two set degenerates outright. We therefore record exactness for $k \geq 6$ as **open**, with the evidence leaning negative, rather than as a route in progress. The remaining candidate is a computer-assisted existence proof (interval Newton) for an irrational certificate, which the ceiling makes tractable in principle, since certifying the equation $G(w) = G^*$ exactly delivers the nullity as a discrete consequence.

How many n does this reach?

The construction needs a divisor d of n with

$$\lceil k/2 \rceil \leq d \leq \frac{n}{2k+4},$$

the left inequality from the rank law and the right from requiring $\Gamma_{n/d}$ to contain $k+1$ distinct interior values. This is a real restriction and it cuts both ways.

Proposition 41 (Reachability). *The set of n admitting such a divisor has density 1, but its complement contains every prime, and also every $n = 2p$ once $k \geq 6$ (since then $d \geq 3$ while the only small divisor is 2). Convergence is slow: at $k = 6$ the reachable proportion of n is 54% up to 240, 71% up to 1000 and 79% up to 5000; at $k = 8$, 36%, 62% and 73%.*

In particular **prime n lies outside the period- d method entirely**, because d must divide n and $d = 1$ gives rank 3. That is not a gap in the search but a structural feature: for prime n the only available construction is the fully non-equivariant one, i.e. solving $T = I$ directly in the $5n$ weights, which is the regime where our searches plateau. Since $Z(P(23, 5)) = 12 = 2k + 2$ with 23 prime, such n certainly can attain the ceiling; we simply cannot certify them this way.

Larger d is not merely permitted but helpful. The rank law requires only $d \geq \lceil k/2 \rceil$, and taking d strictly larger enlarges the solution manifold from dimension $5d - (k+1)$, giving more room to avoid the degenerate locus. The effect is visible at the tightest configurations. At $k = 6$, $d = 3$, $n = 48$ the grid Γ_{16} offers exactly 7 interior values for the 7 required — a unique root set, no slack — and every solution collapses a spoke. At $k = 6$, $d = 4$, $n = 64$ the grid is again Γ_{16} with the same unique root set, but now the manifold has dimension $20 - 7 = 13$ rather than $15 - 7 = 8$, and a realisable certificate is found on the first attempt (smallest required entry 0.086, gap 6.8×10^{-2}). So the degeneracy of the previous subsection is a symptom of working at the minimum period, not an obstruction inherent to tight root sets. Certificates were likewise found at $k = 6$, $d = 6$, $n = 96$.

Within the reachable set, the construction appears to work essentially always. Sweeping n for $k = 6$ at $d = 3$ produced a realisable certificate at *every* n tested:

$$n = 60, 72, 84, 90, 96, 108, 120,$$

most of them on the first root set tried, with smallest required entries between 0.12 and 0.46. So

$$Z(P(n, 6)) = M(P(n, 6)) = 14 \quad \text{for each of these } n,$$

and the earlier appearance of a scattered list of congruence classes was an artefact of the cyclotomic route, not of the problem. We state this as a computation rather than a theorem: we have no proof that a realisable root set exists for every reachable n , only that one was found in every case examined.

Remark 42 (Status of Theorem 39). Unlike the cyclotomic certificates of Section 8, which are integer matrices whose nullity is confirmed by exact rational elimination, these are found by a Newton solve and their weights are algebraic numbers of no recognised form. We therefore record Theorem 39 as **numerically certified but not yet verified in exact arithmetic**. The evidence is strong — a thirteen-order spectral separation, the correct zero pattern with every required entry a fifth of the matrix scale, and the nullity never exceeding the ceiling $2k + 2$, which is an automatic check on the whole pipeline — but it is not of the same kind as the $k = 4$ and $k = 5$ results, and we do not claim that it is. The natural route to closing the gap is to choose the root set to be a union of full Galois orbits of Γ_m , which makes G^* a product of the Ψ_e and hence puts the system $G(w) = G^*$ over $\mathbb{Q}$; that is in progress.

11 Beyond congruence classes: a certified value at a prime

Theorem 37 says the cyclotomic layer cannot produce a value valid for all large n , and points at the replacement: Corollary 12 characterises the ceiling for *every* matrix on the pattern by the condition $T = I$, which carries no arithmetic. This section carries that out at a prime n , where by Theorem 37(i) no period-one certificate can reach.

A better system to certify

Optimising $T = I$ is convenient but bad to certify: the map $w \mapsto T$ has a 51-dimensional image inside the 64-dimensional space of 8×8 matrices at $k = 3$, so $\{T = I\}$ is cut out by 13 local equations we cannot identify and no square subsystem is available. We certify the nullity directly instead.

Now $\text{null } A \geq r$ if and only if some full-rank $K \in \mathbb{R}^{2n \times r}$ has $AK = 0$. Fixing a row permutation P and writing K in graph form $K = P \begin{pmatrix} I_r \\ X \end{pmatrix}$ makes $\text{rank } K = r$ automatic, and

$$f(w, X) := A(w) K(X) = 0$$

is *bilinear*: quadratic of total degree 2, with rational coefficients. Its Jacobian is affine, so the Lipschitz constant of J is an explicit integer, and a Krawczyk contraction bound needs only three norms rather than an interval matrix product.

The choice of which equations to solve is not free. The system $AK = 0$ has $2nr$ equations of rank $2nr - \binom{r}{2}$, and the deficiency is structural: $K^\top AK$ is symmetric because A is, and $\binom{r}{2}$ is exactly its antisymmetric part. Concretely, if the $(2n - r)r$ equations on the non-pivot rows $P[r:]$ vanish then $K^\top AK$ equals the $r \times r$ block $(AK)[P[r:]]$, which must therefore be symmetric, so setting its upper triangle to zero forces the lower triangle as well. Hence

$$\underbrace{(AK)[P[r+m], t] = 0}_{(2n-r)r \text{ equations}} \quad \text{and} \quad \underbrace{(AK)[P[s], t] = 0, s \leq t}_{\binom{r+1}{2} \text{ equations}}$$

is *exactly equivalent* to $AK = 0$. Selecting rows by pivoting instead would leave $\binom{r}{2}$ equations unproved.

Theorem 43 (Certified values off every congruence class). *For each odd n with $17 \leq n \leq 27$,*

$$Z(P(n, 3)) = M(P(n, 3)) = 8.$$

In particular this holds at the primes $n = 17, 19, 23$.

Proof. $Z(P(17, 3)) \leq 8$ is the rotation bootstrap. For the lower bound, take $n = 17$, $k = 3$, $r = 8$, so $N = 34$ and the system above has $85 + 208 = 293$ unknowns and 272 equations, of which $208 + 36 = 244$ are the structural ones. At a numerically located point the Jacobian of the 244 structural equations has rank exactly 244, so fixing the remaining 49 unknowns at explicit rationals with denominator 10^4 leaves a square regular system. Newton in 60-digit arithmetic drives its residual to 3.2×10^{-61} ; the 28 dependent equations, which are not solved for, fall to 4.1×10^{-61} , confirming the structural reduction. With Y an approximate inverse of the square Jacobian,

$$\|Y\|_\infty = 1.54 \times 10^3, \quad \|I - YJ\|_\infty = 2.18 \times 10^{-12}, \quad \|Yf\|_\infty = 2.85 \times 10^{-60},$$

and with $L = 12$ bounding the Lipschitz constant of J in the ∞ -norm, the Krawczyk test on the box of radius $\delta = 10^{-12}$ gives $\alpha = \|I - YJ\|_\infty + \|Y\|_\infty L \delta = 1.85 \times 10^{-8} < 1$ and $\|Yf\|_\infty + \alpha \delta \leq \delta$. So a true zero (w^*, X^*) exists in that box. Then $A(w^*)K(X^*) = 0$ with $\text{rank } K = 8$, so $\text{null } A(w^*) \geq 8$, and ≤ 8 by Corollary 12. Every edge weight of $A(w^*)$ exceeds 0.05 times the matrix scale and the ninth singular value exceeds 0.126 times it, so the box is far from any degenerate stratum and the pattern is exactly that of $P(17, 3)$.

The cases $n = 19, 21, 23, 25, 27$ are identical in structure and each passes the same test; for $n = 19$, to give one more set of numbers, there are 335 unknowns and 304 equations of which 276 are structural and 28 dependent, 59 unknowns are fixed at rationals, and $\|Y\|_\infty = 4.37 \times 10^2$, $\|I - YJ\|_\infty = 1.06 \times 10^{-12}$, $\|Yf\|_\infty = 1.20 \times 10^{-60}$, with Krawczyk passing at $\delta = 10^{-13}$ and $\alpha = 5.25 \times 10^{-10}$. See `verification/krawczyk.py` and `verification/certify_pipeline.py`. $\square$

Together with Theorem 23 this gives $Z(P(n, 3)) = 8$ at $n = 10, 17, 19, 20, 21, 23, 25, 27, 30, 40, \dots$ — a set that no single divisibility condition describes. The construction fails at $n = 13$ and $n = 15$, where by (3) the solution variety has dimension only 3 and 9; it succeeds from $n = 17$, where the dimension is 15. We have not tested even n in this range, since the gauge fixing used here is cleanest for n odd.

Remark 44 (Three independent numerical routes, one certificate). The certificate at $n = 17$ was located three separate ways — by minimising $\|T - I\|$ over all $5n$ weights, by an alternating solve of $AK = 0$ with the spokes normalised, and by the fully gauge-fixed search of (3) — and all three lead to a Krawczyk-certified matrix of nullity 8, with the gauge-fixed route giving the best conditioning ($\|Y\|_\infty = 2.1 \times 10^2$, ninth singular value 0.167 of the matrix scale, minimum edge weight 0.146). Since stage 1 is only a search and stages 2–3 are a proof, the numerical route taken does not enter the result; that three unrelated ones agree is a check on the search, not on the certificate.

Since 17 and 19 are prime, Theorem 37(i) shows no period-one certificate — in particular nothing in Sections 8 or 10 — can give these values. Indeed no divisibility class containing either of them is available at all: a class $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only when L is that prime, and Γ_{17} offers only Ψ_1 and Ψ_{17} , the latter of degree $8 \neq k + 1 = 4$. These are the first values in this paper that do not sit on a divisibility class.

The reachable set stabilises

That $n = 17$ works and $n = 13$ does not is explained by a measurement rather than by arithmetic. The dimension of the reachable set of monodromies, that is the rank of $\partial T / \partial w$, grows with n and then *saturates*:

$k = 2, n$	5	6	7	8	9	10	11	12	≥ 13
rank	19	23	26	28	30	31	31	31	31
$k = 3, n$	7	9	11	13	14	15	16	17	≥ 18
rank	27	34	40	46	49	51	51	51	51

Past $n = 11$ for $k = 2$ and $n = 15$ for $k = 3$, increasing n adds no new reachable monodromies. So whether I is reachable stops depending on n in any essential way once n passes roughly $4k + 3$, and the observed pattern of successes and failures follows it: for $k = 3$ the ceiling is attained at $n = 17, 20$ and not at $n = 13 < 15$. Independently, at $n = 17$, $k = 3$ the solution variety of $f = 0$ has dimension $293 - 244 = 49 = 5n - \binom{2k+3}{2}$, exactly the codimension count for symmetric matrices on the pattern with nullity $2k + 2$.

There is also a hard *necessary* condition, from spending the gauge. The conjugation $A \mapsto DAD$ by a positive diagonal matrix has $2n$ parameters and preserves symmetry, the pattern and the nullity; the n inner parameters normalise every spoke to 1, and for odd n the n outer ones normalise every outer weight to ± 1 , since $u_i + u_{i+1} = -\log |b_i|$ is uniquely solvable on an odd cycle. (The product of the signs around the cycle is a gauge invariant, so there are exactly two slices.) What remains is $2n$ diagonal entries and n inner weights, with no residual gauge, and the solution variety of $f = 0$ has dimension

$$[3n + (2n - r)r] - [(2n - r)r + \binom{r+1}{2}] = 3n - (k + 1)(2k + 3),$$

which is nonnegative only when

$$n \geq \frac{(k + 1)(2k + 3)}{3} = 7, 12, 19, 26, 35, 46, 57 \quad (k = 2, \dots, 8). \quad (3)$$

Conjecture 45. *For every $k \geq 2$ there is $N(k)$ with $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all $n \geq N(k)$.*

For $k = 2$ this is Theorem 20, with $N(2) = 11$, proved. We do not conjecture that $N(k)$ is linear: (3) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic, and at $k = 6$ that bound is 35 — larger than the value $4k + 3 = 27$ that the saturation data alone might suggest. The two quantities measure different things, and the dimension count is the binding one. Our data is consistent with $N(k)$ exceeding (3) by a small additive amount: $N(2) = 11$ against a bound of 7, and $14 \leq N(3) \leq 17$ against a bound of 12, since the ceiling is attained at $n = 17$ and 20 and not at $n = 13$.

The content of the conjecture for $k \geq 3$ is a reachability statement about products of transfer matrices, with no arithmetic in it — which is why we expect it to behave uniformly in n , unlike everything in Sections 8 and 10.

Remark 46 (A strict gap, and a correction to our own evidence). Two caveats. First, $n = 10$, $k = 2$ resists the search even though its reachable dimension is already saturated, and $Z(P(10, 2)) = 6$; so we expect $M(P(10, 2)) < Z(P(10, 2))$, a strict gap, though we have not proved it. Second, the eigenvalue-minimising search we reported earlier is unreliable and its negative results should not be read as evidence: it gave maximum nullity 7 at $P(20, 3)$, whereas the ceiling 8 is attained there both by Theorem 23 and, in the *symmetric* class, by the method of this section. It was collapsing onto the degenerate strata — all spokes zero, or all edges zero — that the formulations here exclude by construction.

12 Why lower bounds resist standard methods

A set is zero forcing iff it meets every *fort* — a nonempty F such that every vertex outside F has 0 or ≥ 2 neighbours in F — so Z is a minimum hitting set over forts. Three standard consequences all fail here, and all degrade as k grows.

- **Disjoint forts.** Pairwise disjoint forts $F_1, \dots, F_t$ force $Z \geq t$. But forts in $P(n, k)$ have 7–13 vertices while the graph has $2n \leq 56$, and a greedy search found at most *two* disjoint forts in every case tested: the certificate cannot exist at these sizes.
- **The fort LP.** Computing the exact LP value by cutting planes gives $\text{LP}/Z \approx 0.43\text{--}0.50$, e.g. $\text{LP} = 3.00$ against $Z = 7$ for $P(12, 3)$, and 3.21 against $Z = 10$ for $P(18, 4)$. The integrality gap is about 2 and widens with k . This also explains why fort-based exact solvers converge slowly on this family: branch and bound must close a factor-2 gap by branching.
- **Spectral.** The Hoffman-type bound $Z \geq N(1 + 2\lambda_{\min}/(d - \lambda_{\min}))$ yields values in 1.2–2.9 against true Z of 6–12, and also weakens as k grows.

13 Status of the claims

- **Proved:** Theorems 3 and 7 and the corollary, for all n in the stated ranges; Theorem 11, Corollary 12, Propositions 13 and 14, and Theorem 28.
- **Proved (exact certificates):** Theorem 34, giving $Z(P(n, 3)) = 8$ for $n = 40, 60, 80, 120$; $Z(P(n, 4)) = 10$ for $n = 60, 70, 90, 120$ and for every n divisible by 60, 70 or 90; $Z(P(n, 5)) = 12$ for every n divisible by 24; $Z(P(n, 2)) = 6$ for $9 \leq n \leq 130$, $n \neq 10$. Each rests on an explicit matrix, integral in the cyclotomic cases and otherwise verified in $\mathbb{Q}(\zeta_n)$, together with Theorem 3. These are the first exact values for $k \geq 4$.
- **Proved (finite):** every tabulated value of Z , by exhaustive search.
- **Numerically certified, not yet exact:** Theorem 39, $Z(P(60, 6)) = Z(P(72, 6)) = 14$, $Z(P(96, 7)) = 16$, $Z(P(96, 8)) = 18$ — the first values for $k \geq 6$. The certificates are period-three matrices whose weights are algebraic numbers from a Newton solve, verified on the full $2n \times 2n$ matrix with a thirteen-order spectral separation and every required entry a fifth of the matrix scale, but not in exact arithmetic. We do not claim these at the same standard as the $k = 4, 5$ results.
- **Observation supported by computation:** the rank law $\text{rank} = \min(2d+1, k+1)$, verified for $3 \leq k \leq 13$ and $1 \leq d \leq 7$, and returning the known value 3 at $d = 1$. Not proved.
- **Conjectural:** Conjecture 8. The sequences are non-monotone, so extrapolation is not safe: five consecutive equal values occurred at $k = 5$ ($Z = 10$ for $17 \leq n \leq 21$) before the sequence moved to 11 and then 12, and at $k = 6$ the bound is attained at $n = 28$ and lost at $n = 29$.
- **Observation, not a theorem:** that the obstruction to nullity $2k + 2$ at the remaining n is arithmetic rather than dimensional. The parameter budgets are met, and extensive search in both matrix classes plateaus well above machine precision; but no proof is offered, and no (n, k) is proved to have $M(P(n, k)) < 2k + 2$.
- **Refuted (ours):** that the period-one construction “never exceeds 6, for every $k \geq 2$ ”. Three roots can be prescribed and no more, but the remaining $k - 2$ roots can land on the grid by Galois conjugacy, and do: period-one certificates reach $2k + 2$ for $k = 2, 3, 4, 5$. The period-two construction we built to get past 6 was unnecessary, and the values it gave (8 for $P(n, 4)$, 10 for $P(24, 6)$) are superseded.

- **Refuted (ours):** the single-threshold form $n_0(k) = 5k - 2$, by $Z(P(29, 6)) = 12$. It was recorded as a falsifiable prediction and it failed.
- **Refuted (ours):** any bound of the form $Z \leq 2(\gcd(n, j) + \gcd(n, k))$, by $Z(P(11, 3)) = 7 > 4$ and five similar cases. Observation 9 survives only as a statement about $d_1, d_2 > 1$.
- **Open:** whether $Z(P(n, k)) = 2k + 2$ for *all* large n — note that by Proposition 41 the period- d method cannot settle prime n at all, so a different construction is needed there — and whether $M(P(n, k)) = Z(P(n, k))$ in general. Also open is whether the degeneracy of Section 10 is essential: at $n = 48$ and $n = 54$, where the choice of root set has no slack, every solution collapses a spoke, and we have no proof that this must happen rather than merely observing that it does. By Corollary 12 the certificate method could in principle settle the first question completely, since its ceiling is exactly $2k + 2$; what is missing is a construction valid for every residue class of n , and for $k \geq 6$ any construction at all. The three combinatorial techniques of Section 12 are insufficient in principle rather than merely in practice.
- **Not determined:** whether Theorem 7 is attained for $k \geq 3$ (it is attained at $k = 1, 2$). Theorem 3 is attained at $j = 1$ (via the corollary) and at $I(18, 2, 3)$, but is far from sharp elsewhere in the regime where it carries new content; see Remark 6.

References

- [1] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, Computing the zero forcing number for generalized Petersen graphs, *J. Algebra Combin. Discrete Struct. Appl.* 7(2) (2020), 183–193.
- [2] A. Krishnan, A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$, arXiv:2607.19412 (2026).
- [3] J. H. Conway and A. J. Jones, Trigonometric diophantine equations (On vanishing sums of roots of unity), *Acta Arith.* 30 (1976), 229–240.
- [4] I. Niven, *Irrational Numbers*, Carus Mathematical Monographs 11, Mathematical Association of America, 1956, Corollary 3.12.